

CICFlowmeter has 84 features while NFStream by default has a total of 86 out of which 48 match CICFlowmeter's features.

1. CICFlowMeter Features Natively Present in NFStream

CICFlowMeter Name	NFStream Feature	Definition
Flow Duration	bidirectional_duration_ms	The total duration of the network flow in milliseconds.
Tot Fwd Pkts / Tot Bwd Pkts	src2dst_packets / dst2src_packets	Total number of packets sent in the forward (source to destination) and backward directions.
TotLen Fwd Pkts / TotLen Bwd Pkts	src2dst_bytes / dst2src_bytes	Total application payload bytes sent in the forward and backward directions.
Fwd/Bwd Pkt Len (Min/Max/Mean/Std)	[direction]_[min/max/mean/stddev]_ps	Statistical distribution metrics of the application payload sizes.
Flow/Fwd/Bwd IAT (Min/Max/Mean/Std)	[direction]_[min/max/mean/stddev]_piat_ms	Packet Inter-Arrival Time statistics natively computed in C for high performance.
[Flag] Flag Cnt (SYN, ACK, FIN, etc.)	bidirectional_[flag]_packets	Accumulators counting the occurrences of specific TCP flags within the flow.

2. Features Derivable via Post-Processing & Custom Plugins

The remaining 36 features, about 11, can be calculated simply using pandas after processing the pcap. The rest 24 features can be created using custom NFplugins.

CICFlowMeter Name	Implementation Method	Definition
Flow Byts/s & Flow Pkts/s	Pandas Math	Total payload bytes and packets divided by flow duration in seconds.
Down/Up Ratio	Pandas Math	Ratio of backward to forward packets, calculated using integer division (/) to perfectly mirror CICFlowMeter's Java casting logic.
Pkt Len Var	Pandas Math	Calculated by squaring the native payload standard deviation metric.
Fwd/Bwd Header Len	HeaderLenPlugin	Cumulative sum of IP and Transport layer header bytes tracked iteratively.

CICFlowMeter Name	Implementation Method	Definition
Init Fwd/Bwd Win Byts	InitWindowPlugin	The initial TCP window size requested, extracted via byte-array parsing on the initial SYN and SYN-ACK packets.
Active/Idle Time Metrics (Active Mean/Min/Std/Max, Idle Mean/Min/Std/Max)	ActiveIdlePlugin	Statistics on periods of continuous data transmission (Active) versus periods of silence exceeding 5000ms (Idle), tracked via Welford's algorithm.
Bulk Transfer Metrics (Fwd Byts/b Avg,Fwd Pkts/b Avg,Bwd Byts/b Avg,Bwd Pkts/b Avg,Fwd Blk Rate Avg,Bwd Blk Rate Avg)	BulkPlugin	Identifies rapid, continuous data transmission phases exceeding a 4-packet threshold, capturing average byte/packet rates.
Fwd Act Data Pkts & Fwd Seg Size Min	ExtraFeaturesPlugin	Tracks the count of packets containing actual payloads (ignoring empty ACKs) and the absolute minimum header size observed.

3. Extra Features in NFStream (Deep Packet Inspection)

By moving to the NFStream architecture, the dataset natively benefits from nDPI engine integration. This provides high-value Layer 7 protocol visibility and cryptographic context critical for advanced threat detection models. The unique 38 features are: -

NFStream Feature Name	Definition / Value for Cybersecurity Analysis
src_mac	The MAC address of the source device. Highly useful for tracking specific physical devices within a local network (LAN) regardless of IP address changes.
src_oui	The Organizationally Unique Identifier (OUI) of the source MAC. Identifies the hardware vendor (e.g., Apple, Cisco), which helps profile device types (like IoT vs. Workstations).
dst_mac	The MAC address of the destination device.
dst_oui	The Organizationally Unique Identifier (OUI) of the destination MAC.
vlan_id	The Virtual LAN (VLAN) identifier. Crucial for understanding network segmentation and detecting lateral movement across internal network boundaries.
ip_version	The Internet Protocol version (4 or 6).

	Useful for isolating traffic types, as some modern malware specifically leverages IPv6 to bypass legacy firewalls.
tunnel_id	Identifier for tunneling protocols (e.g., VXLAN, GRE). Important for detecting hidden or encapsulated overlay traffic.
application_name	The specific application protocol identified by the nDPI engine (e.g., HTTP, TLS, DNS, SSH, BitTorrent). Critical for identifying anomalous traffic on non-standard ports.
application_category_name	The broader category of the detected application (e.g., Web, Network, Media, VPN). Great for macro-level anomaly detection.
application_is_guessed	A boolean (0 or 1) indicating if the DPI engine had to guess the protocol via heuristics rather than a definitive payload signature match.
application_confidence	A numerical score representing the reliability and depth of the DPI classification (e.g., 6 implies definitive Deep Packet Inspection). Validates the trust level of the dataset labels.
requested_server_name	The domain name requested by the client, typically extracted from raw DNS queries or the TLS Server Name Indication (SNI). Essential for detecting DGA (Domain Generation Algorithm)

	malware communication.
client_fingerprint	Cryptographic client hashes, such as JA3/JA4 (TLS) or HASSH (SSH). Highly effective for profiling specific client software or identifying automated malware/C2 beacons that reuse known cryptographic library configurations.
server_fingerprint	Cryptographic server hashes (e.g., JA3S, HASSHServer) used to identify specific, potentially malicious, server configurations.
user_agent	The extracted HTTP User-Agent string or QUIC User Agent Identifier. Useful for identifying non-standard, spoofed, or anomalous automated tools executing HTTP requests.
content_type	The HTTP Content-Type (e.g., application/json, text/html, application/octet-stream). Helps identify unexpected payload transfers, such as downloading executables from a web page.
expiration_id	A code indicating exactly why the flow was terminated and exported by NFStream (e.g., 0 for idle timeout, naturally closed via FIN/RST flags, etc.). Helps distinguish between naturally completed flows and timed-out connections.